

OligoTox Open Data Challenge — Phase 2

Narrative Document

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Submission target:	Single PDF, ≤12 pages, 8.5"×11", ≥1" margins, ≥11pt Arial, line spacing ≥1.0
Challenge:	NIH NCATS Oligonucleotide Toxicity (OligoTox) Open Data Challenge — Phase 2 (Data Generation)

STATUS: DRAFT. This document is structured to match the six required narrative sections exactly. Sections 1, 2, and 4 are placeholders — they require the actual generated dataset, which does not exist yet, and must not be filled with invented numbers. Sections 3, 5, and 6 describe the experimental design, the gap this work addresses, and the modeling plan — none of that content depends on results, so it is written in full below and should only need light editing once real data collection begins (e.g., to reflect deviations from the plan). **Do not submit until Sections 1, 2, and 4 are rewritten from real experimental outcomes.**

1. Executive Summary of the Dataset(s) Generated, and Positive/Negative Controls Included

[PLACEHOLDER — requires the completed dataset.] When filled in, this section reports:

- The final oligo count actually synthesized and screened, broken out by chemistry class
- Which toxicity domains from the priority list were actually covered (hepatotoxicity, nephrotoxicity, immunotoxicity, complement activation, coagulopathy, thrombocytopenia)
- Which of the in vitro human systems described in Section 3 were actually run
- The positive/negative controls actually included (see the candidate list in 02_methodology_document.md §5) and a one-line statement of whether each behaved as expected

2. Summary of the Main Findings and Conclusions

[PLACEHOLDER — requires real assay results.] When filled in, this section reports actual outcomes: which chemistries or sequence motifs produced the strongest toxicity signal in which system; whether control compounds validated the assay panel (Z'-factor, CV, direction of effect matching literature expectation); any unexpected or negative findings (both are informative for an open dataset); and summary statistics such as the fraction of the library

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flagged toxic per endpoint and the proportion of dose-response curves meeting the fit-quality threshold.

3. Description of How Data Were Produced

3.1 Oligonucleotide Library

The library is designed to give the resulting dataset broad, systematic coverage of the chemical variables believed to drive oligo toxicity, rather than being a random or convenience sample. It spans:

- **Backbone chemistries:** phosphorothioate (PS), phosphodiester (PO), phosphorodiamidate morpholino (PMO), LNA-mixed backbones, peptide nucleic acid (PNA), and 2'-F-ANA
- **Sugar modifications:** unmodified DNA, 2'-O-methyl (2'-OMe), 2'-fluoro (2'-F), locked nucleic acid (LNA), and mixed-modification patterns
- **Structural classes:** gapmers (with systematic wing/gap length variation), siRNA/siRNA-like duplexes, aptamer scaffolds, splice-switching oligonucleotides, and GalNAc- or lipid-conjugated ASOs
- **Systematic gradients:** CpG-dinucleotide content, overall GC content, and sequence length, each varied independently of chemistry so their individual contribution to toxicity signal can be isolated statistically rather than confounded with backbone choice
- **Controls:** clinically approved/advanced reference ASOs with known safety profiles (as benchmarks), known toxicity-positive compounds (e.g., a CpG ODN TLR9 agonist, a hepatotoxic ASO analog), and scrambled/non-targeting sequences matched in chemistry (as chemistry-matched negative controls) — full rationale for each in 02_methodology_document.md

This design choice — varying one axis (chemistry, CpG content, GC content, length) at a time against otherwise-matched backgrounds — is what lets the resulting dataset support attribution (which variable actually drove an observed toxicity signal), not just prediction.

3.2 In Vitro Human Model Systems

Toxicity is assessed using human-derived in vitro systems chosen for direct relevance to the priority toxicity domains, rather than substitute animal models:

System	Toxicity domain	Why this system
Primary human hepatocytes (PHH), sandwich culture	Hepatotoxicity	Retains CYP activity, transport protein expression, and hepatocyte polarity relevant to oligo uptake/metabolism — the field's accepted gold standard in vitro hepatotoxicity model

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System	Toxicity domain	Why this system
Human kidney proximal tubule organoids	Nephrotoxicity	Proximal tubule is the primary site of oligo accumulation via scavenger-receptor uptake and the dominant nephrotoxicity signal in clinical ASO experience; organoids better capture tubular transporter expression than 2D culture
Human PBMC co-culture	Immunotoxicity, complement activation	Captures innate immune activation (monocyte, NK, B-cell) most relevant to CpG- and PS-backbone-driven immunostimulation
Human platelet-rich plasma / citrated plasma	Thrombocytopenia, coagulopathy	Directly measures platelet activation/aggregation and coagulation factor interaction, the proposed mechanism behind PS-backbone-associated thrombocytopenia

Each system's dosing range, timepoints, and full readout panel are specified in `02_methodology_document.md`. Compounds are prioritized across systems using a tiered strategy (a full panel for a subset of high-information compounds, a narrower panel for the rest), so that the completed dataset trades comprehensive multi-endpoint data on some compounds against broader single-endpoint coverage across the whole library — both are useful signal for model training, and the schema (`dataset/data_dictionary_and_schema.md`) records exactly which combinations were tested and which were not.

3.3 Data Acquisition and Computational Processing

Data acquisition uses automated liquid handling for compound dosing and plate-reader-based endpoint capture (luminescence, fluorescence, absorbance, multiplex immunoassay), supplemented by transcriptomic and/or proteomic profiling for a subset of high-priority compounds.

Raw instrument output is processed as follows before entering the released dataset:

1. Normalization of each raw reading to its plate's vehicle control
2. Per-plate quality control: Z'-factor (target ≥ 0.5) and vehicle-control coefficient of variation (target $\leq 15\%$) computed from positive/negative control wells; plates failing either threshold are flagged and repeated rather than silently included
3. Outlier detection across biological replicates (Grubbs' test)
4. Dose-response curve fitting (4-parameter logistic) on QC-passed, outlier-cleaned data, to derive IC50/EC50 with 95% confidence interval, Emax, Hill coefficient, no-observed-effect

concentration (NOEC), and area-under-the-dose-response-curve (AUDRC) as a single summary statistic

5. Batch-level drift correction using repeated anchor compounds present in every batch

Every stage above — raw value, normalized value, QC flag, and derived statistic — is retained in the released dataset (see `dataset/data_dictionary_and_schema.md`, Tables 3–4), not just the final fitted values, so downstream users can re-derive or re-QC the data under their own criteria rather than trusting ours by default.

[Once real data exists: replace the generic description above with the actual number of batches run, actual QC pass/fail rates, and any deviations from this plan — e.g., a system that was dropped, a timepoint that was changed.]

4. Description of How Indicators and Predictor Variables Were Measured, Their Distribution, and the Distribution of Predictor Variables Amongst Tested Oligos

[PLACEHOLDER — requires the actual assembled dataset.] The full definitions of every indicator (toxicity readout) and predictor variable (sequence/chemistry feature) are fixed in advance in `dataset/data_dictionary_and_schema.md` — Table 2 (computed sequence/chemistry features) and Table 3 (experimental readouts). Once real data exists, this section must report: the distribution of each indicator across the tested library (e.g., what fraction of compounds show a dose-dependent hepatotoxicity signal, the spread of IC50 values by endpoint); the distribution of predictor variables actually achieved (how many tested oligos fall into each backbone class, sugar modification, length bucket, GC-content bin, and CpG-count bin — i.e., whether the library actually achieved the systematic chemical diversity described in Section 3.1, or whether gaps emerged during real synthesis/screening); and the coverage matrix of which oligo × assay-system × endpoint combinations have data versus were not tested (per the tiered strategy in Section 3.2).

5. Discussion of How the Results Address a Gap in the Publicly Available Data

Public data available today for training oligo toxicity models has five well-documented limitations, each of which this dataset is designed to directly address:

1. **Small scale.** Most published oligo toxicology datasets contain well under 200 compounds with a single toxicity readout each, too little signal for multi-endpoint or deep-learning approaches. This dataset's systematic library design (Section 3.1) is sized specifically to exceed that scale within a single coordinated study.
2. **Chemical homogeneity.** Public data heavily over-represents phosphorothioate-backbone ASOs; gapmer, LNA, 2'-OMe, and PMO chemistries are

comparatively underrepresented, which biases any model trained only on public data toward PS-backbone chemistry. The library's backbone and sugar-modification axes (Section 3.1) are explicitly designed to correct this imbalance.

3. **Animal origin.** The majority of available toxicology data derives from rodent in vivo studies, which are known to mis-predict human hepatotoxicity and immunotoxicity due to species differences in innate immune receptor profiles and liver biology — the exact motivation stated in NCATS's RFI on this challenge. Every system used here (Section 3.2) is human-derived.
4. **Proprietary lock-up.** The largest existing oligo toxicity datasets are held by industry and not publicly accessible, so the research community cannot use them to build or validate models regardless of their scale or quality. This dataset is released under CC BY 4.0 with no access restriction (see 03_public_access_and_dissemination_plan.md).
5. **Siloed, single-endpoint measurement.** Existing datasets rarely combine orthogonal mechanistic readouts (e.g., transcriptomic + phenotypic) for the same compound, which prevents multi-task modeling that could exploit shared biology across toxicity domains. This dataset's multi-endpoint panel per compound (Section 3.2, tiered strategy) is designed specifically to enable that.

[Once real data exists: replace the general argument above with the actual, quantified gap closed — e.g., "this dataset adds N real human-in-vitro data points across M chemistry classes, a Kx increase over the largest comparable public dataset in [specific toxicity domain]," citing the specific public datasets being compared against.]

6. Discussion of How the Data Could Be Used to Develop a Predictive Model

The dataset is structured, from the schema stage onward, to be directly usable for training and evaluating predictive models of oligo toxicity, in three complementary ways:

Feature-based models. The computed sequence/chemistry features in dataset/data_dictionary_and_schema.md Table 2 (GC content, CpG count, predicted secondary structure, predicted melting temperature, off-target hybridization score, backbone/sugar one-hot encodings, physicochemical properties) support interpretable models such as gradient-boosted trees, where per-prediction feature attribution (e.g., SHAP values) lets a researcher see which molecular features drove a given toxicity prediction — useful for generating testable mechanistic hypotheses, not just point predictions.

Sequence-based / foundation-model approaches. Because the raw nucleotide sequence and per-position chemical modification locations are retained (not just summary chemistry labels), the dataset also supports fine-tuning of pretrained genomic sequence models, which can capture sequence-motif effects (e.g., specific CpG or poly-G contexts) that hand-engineered features may miss.

Multi-task and dose-response-aware modeling. Because each compound is measured across multiple toxicity endpoints where the tiered protocol allows (Section 3.2) and full dose-response curves are retained rather than collapsed to a single binary toxic/non-toxic label (Section 3.3), the dataset supports multi-task learning that can share statistical strength across correlated endpoints, and regression- or uncertainty-aware modeling of the full concentration-response relationship rather than a single threshold call.

Rigorous evaluation. The dataset design also enables benchmark splits harder than a random train/test split: a chemistry-stratified split (train on some backbone/sugar combinations, test on held-out ones, to measure whether a model generalizes across chemistry rather than memorizing it) and — since data collection proceeds in batches (see 02_methodology_document.md) — a prospective split (train on early batches, test on later ones), which is a more honest test of real-world predictive utility than a random split alone.

[Once real data exists: report actual baseline model performance on the real dataset — not simulated numbers — if any modeling has been done as part of the submission, and note which of the above approaches were actually tried.]

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